

13.8 LAB: Digital forensics (Walkthrough)

(Page 1 of 4)

Introduction

Digital forensics is a branch of forensic science focused on recovering, collecting, analyzing, and examining data found in devices. Different forensic techniques include recovering deleting files, analyzing browser activity, and examining metadata.

Conducting forensic analysis first requires a complete copy of a device. The copy must be bit-for-bit to ensure accuracy. The Linux command **dd** is used to make a bit-for-bit copy.

FTK Imager (Forensics Toolkit) is a Windows tool that copies a system similar to **dd**. The toolkit can also analyze emails, decrypt files, and detect malware. **E01** is a forensic image file for a software application called EnCase. Compared to **dd** images, **E01** takes up less disk space and contains metadata.

Learning Objectives

- Create a forensics copy of a system
- View a forensics copy of a system using Autopsy
- Recover and analyze deleted files
- Generate a forensics report

Devices

The lab contains one virtual machine:

- **ZYWIN01** (Windows Server 2022)

Tasks

Task 1: Convert a **dd** image to **E01** using FTK Imager

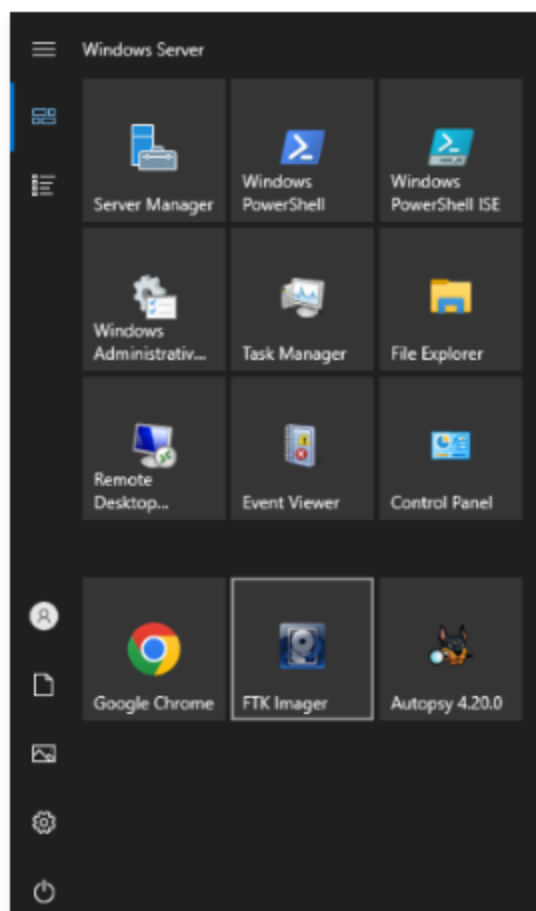
Task 2: Use Autopsy to analyze the file system

Task 3: Generate a report with Autopsy

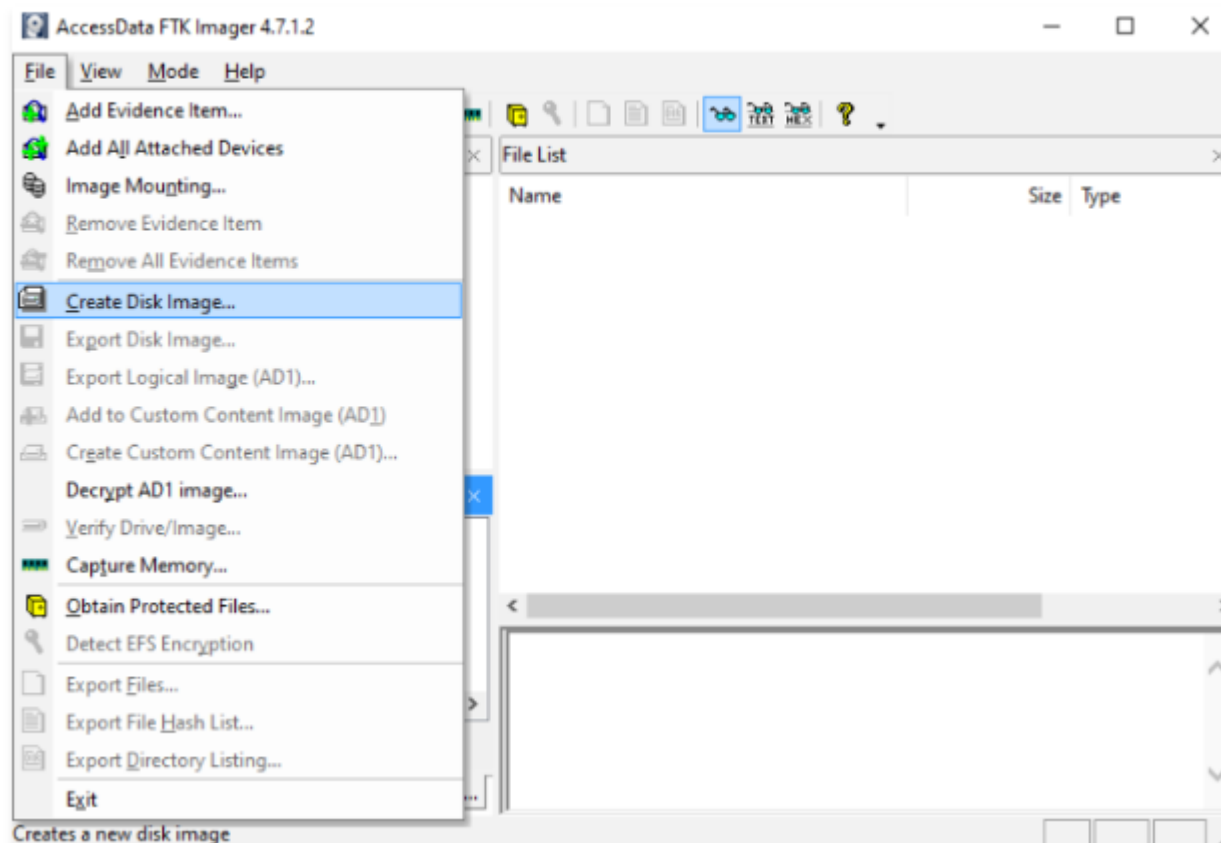
Task 1: Convert a dd image to E01 using FTK Imager

1. Select **ZYWIN01** from the server drop-down menu in the right pane and click **Connect to VM**.

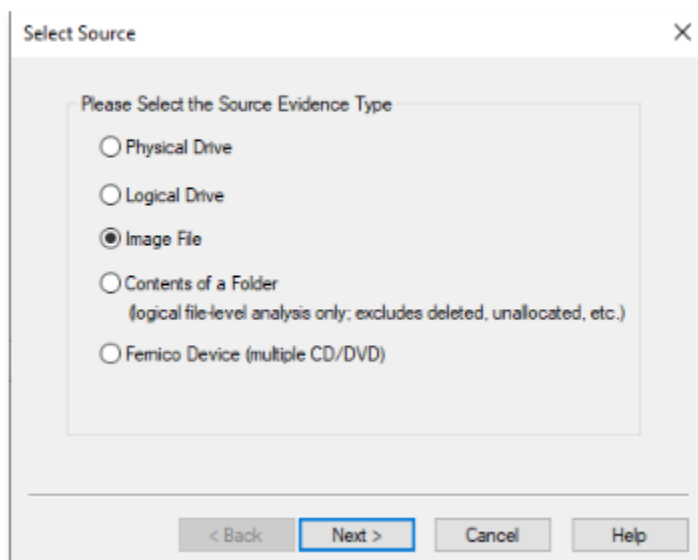
2. Select the **Windows Start Menu** at the bottom left. Select **FTK Imager**.



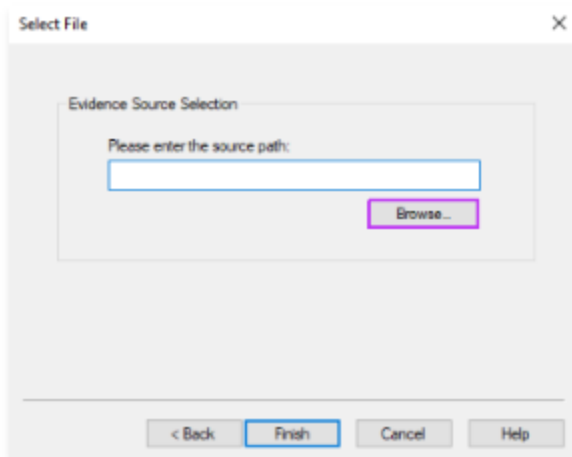
3. Select **File** → **Create Disk Image**.



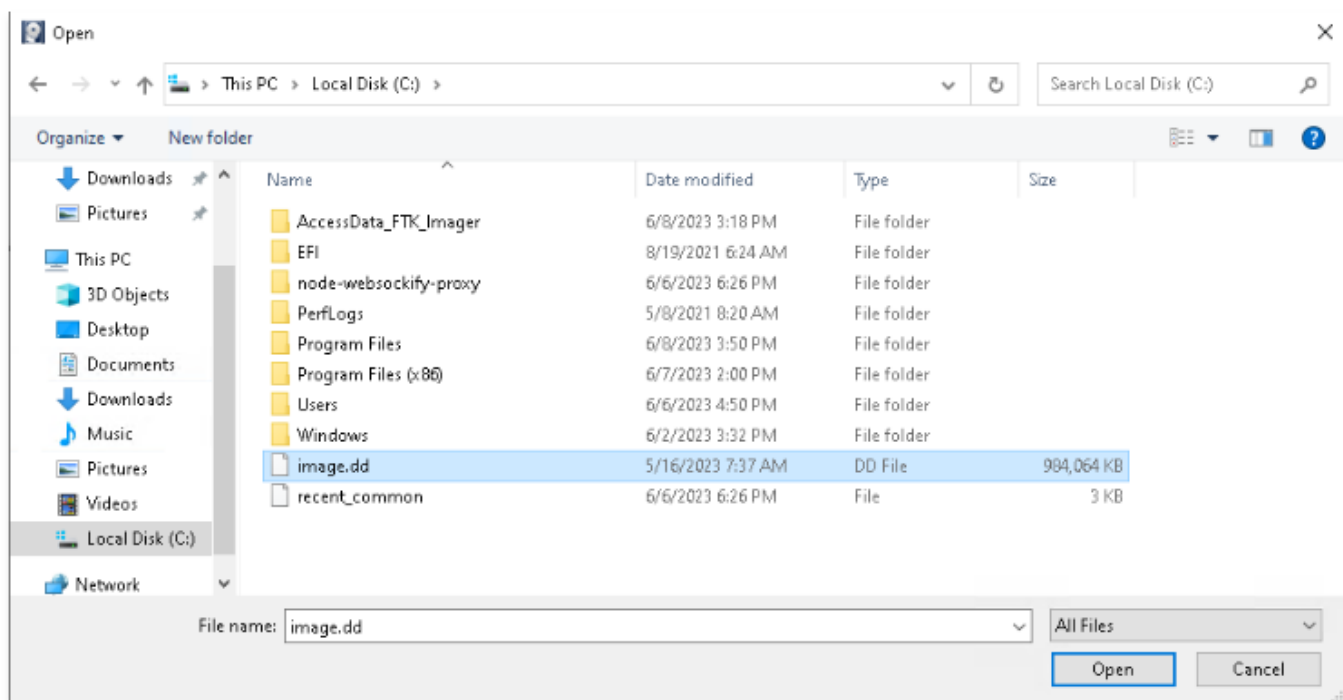
4. Select **Image File** and then **Next**.



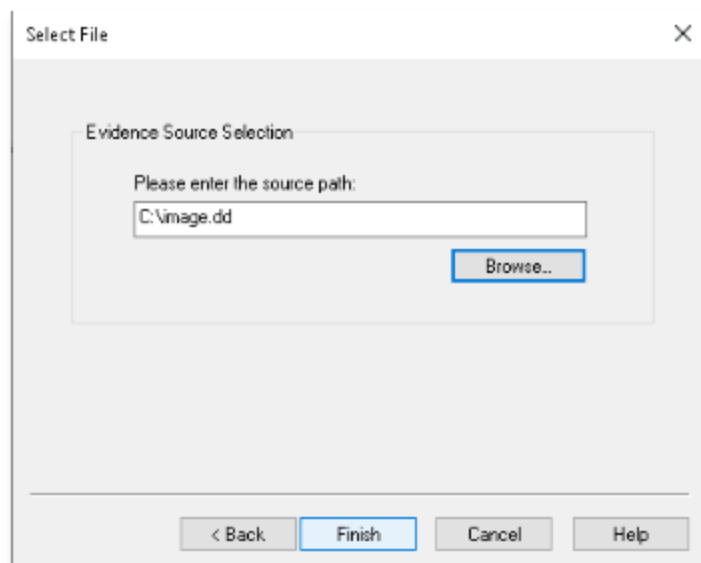
5. Click **Browse**.



6. Go to **This PC** → **Local Disk (C:)**. Select **image.dd** and click **Open**.

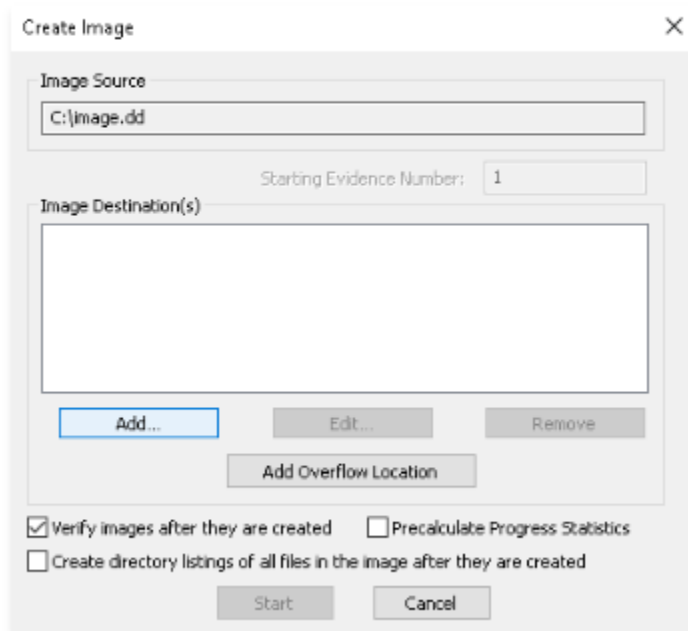


7. Click **Finish**.



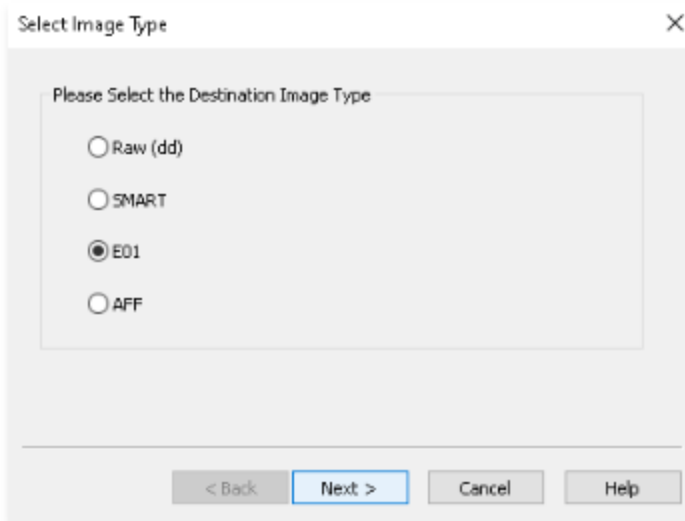
The 'Select File' dialog box has a title bar with a close button (X). Inside, there is a section titled 'Evidence Source Selection'. Below this title, it says 'Please enter the source path:' followed by a text input field containing 'C:\image.dd'. To the right of the input field is a 'Browse...' button. At the bottom of the dialog, there are four buttons: '< Back', 'Finish', 'Cancel', and 'Help'.

8. On **Create Image**, Click **Add**.



The 'Create Image' dialog box has a title bar with a close button (X). It contains several sections: 'Image Source' with a text field containing 'C:\image.dd'; 'Starting Evidence Number:' with a text field containing '1'; 'Image Destination(s)' with a large empty list box. Below the list box are three buttons: 'Add...', 'Edit...', and 'Remove'. Below these is a button labeled 'Add Overflow Location'. At the bottom, there are three checkboxes: 'Verify images after they are created' (checked), 'Precalculate Progress Statistics' (unchecked), and 'Create directory listings of all files in the image after they are created' (unchecked). At the very bottom are 'Start' and 'Cancel' buttons.

9. Choose **E01** to select the destination image type. Click **Next**.



Select Image Type

Please Select the Destination Image Type

☐ Raw (dd)

☐ SMART

☒ E01

☐ AFF

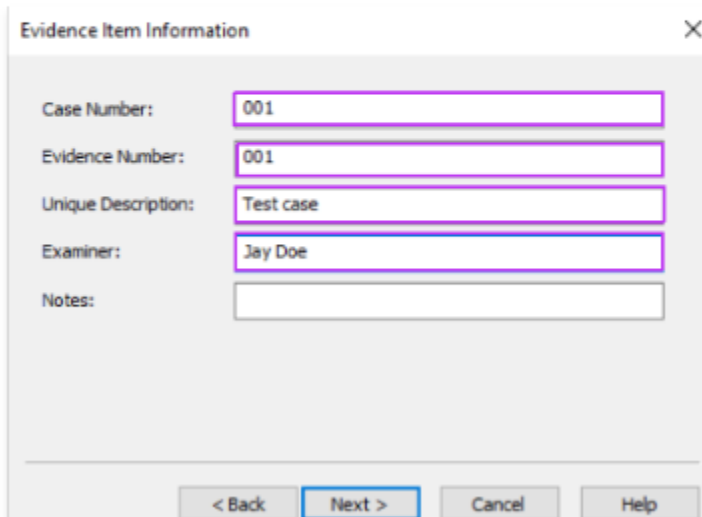
< Back Next > Cancel Help

This dialog box is titled "Select Image Type" and contains a section titled "Please Select the Destination Image Type". It features four radio button options: "Raw (dd)", "SMART", "E01" (which is selected), and "AFF". At the bottom, there are four buttons: "< Back", "Next >" (highlighted with a blue border), "Cancel", and "Help".

10. Enter the following information:

- **Case Number:** 001
- **Evidence Number:** 001
- **Unique Description:** Test case
- **Examiner:** Jay Doe

Click **Next**.



Evidence Item Information

Case Number: 001

Evidence Number: 001

Unique Description: Test case

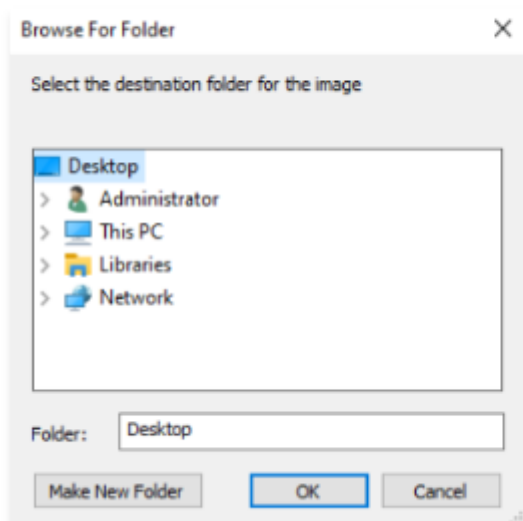
Examiner: Jay Doe

Notes:

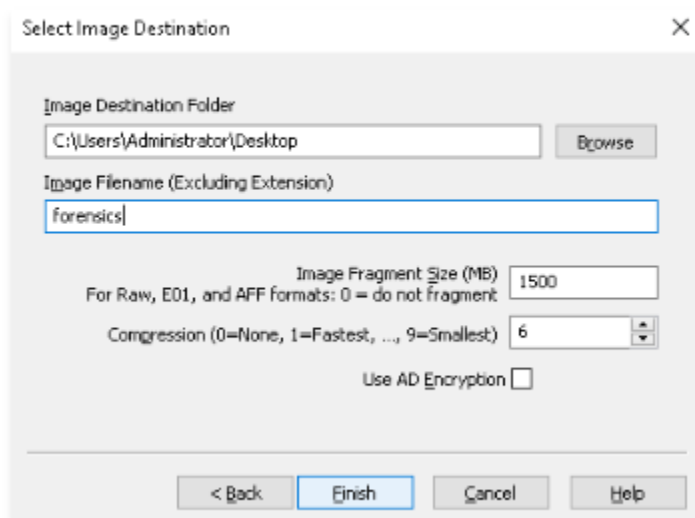
< Back Next > Cancel Help

This dialog box is titled "Evidence Item Information" and contains five input fields. The first four fields are labeled "Case Number:", "Evidence Number:", "Unique Description:", and "Examiner:", and contain the values "001", "001", "Test case", and "Jay Doe" respectively. The fifth field is labeled "Notes:" and is empty. At the bottom, there are four buttons: "< Back", "Next >" (highlighted with a blue border), "Cancel", and "Help".

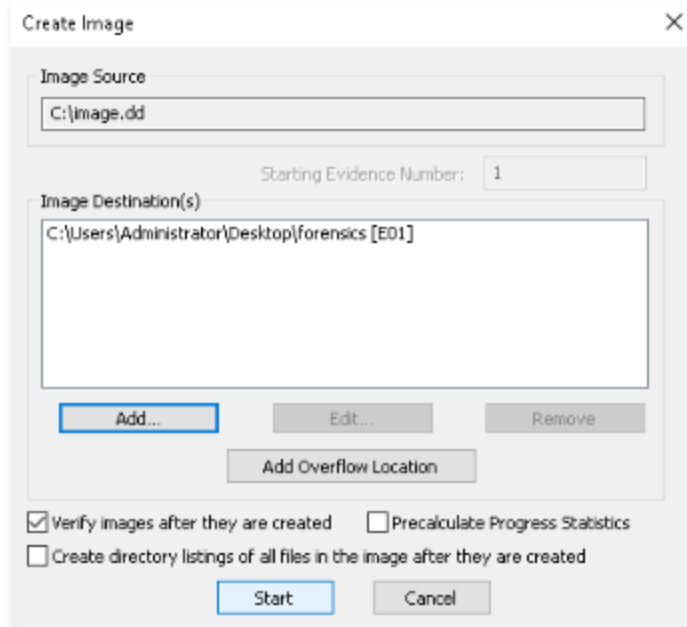
11. For image destination, click **Browse**. Select **Desktop** and click **OK**.



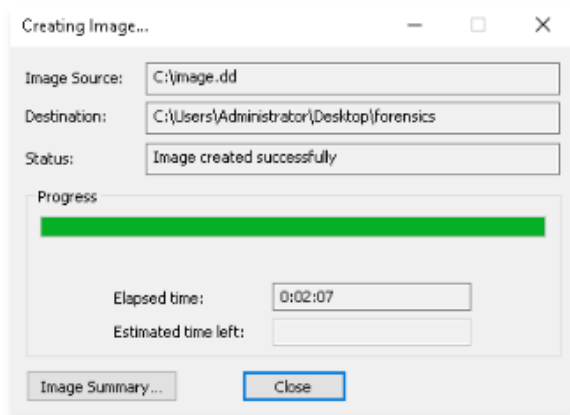
12. Enter **forensics** for **Image Filename** and click **Finish**.



13. Click **Start**.



14. Once the image is created, click **Close** on the **Drive/Image Verify Results** and **Creating Image...** windows. (Note: Image creation may take a few minutes to complete.)



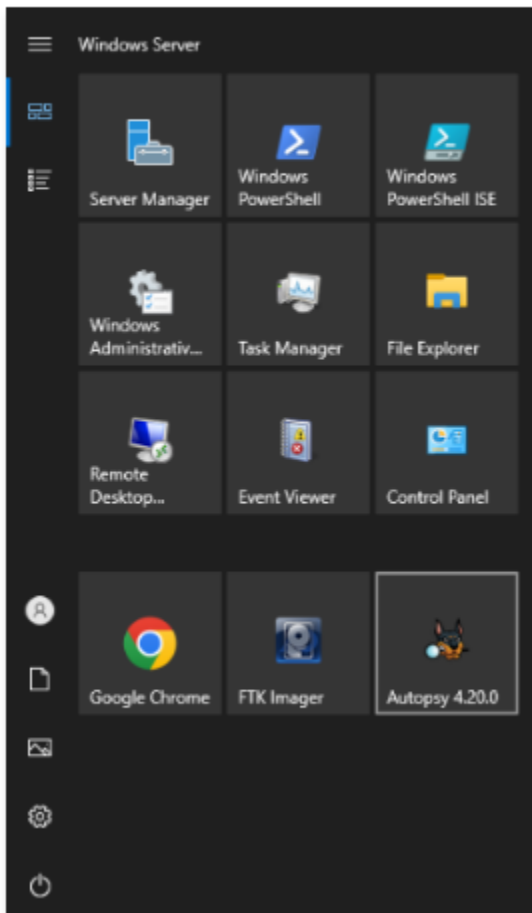
Points

0/3  E01 image created.

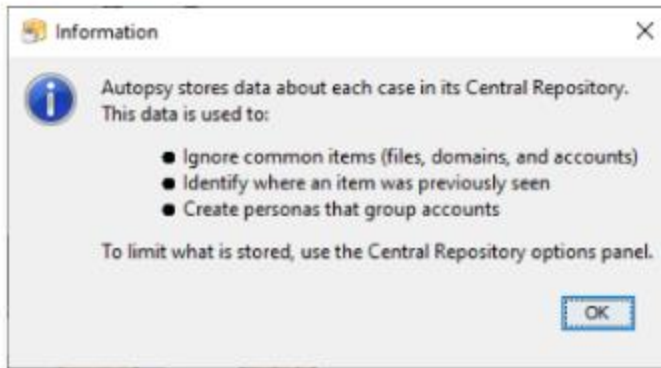
15. **Close** FTK Imager.

Task 2: Use Autopsy to analyze the file system

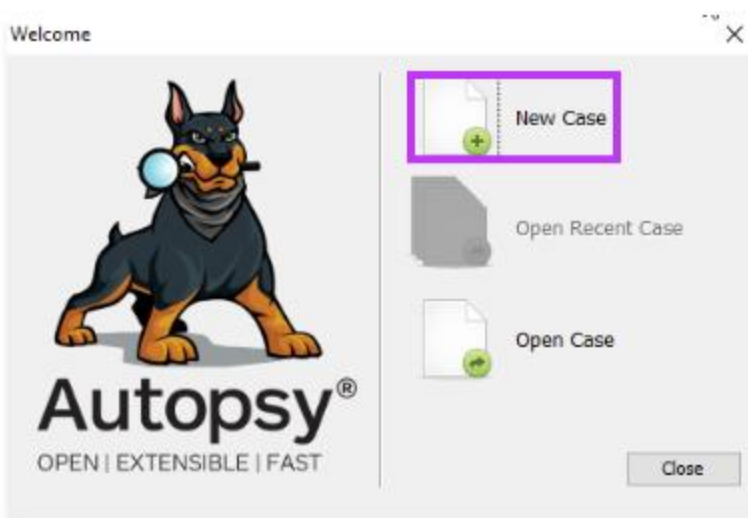
1. Open the **Windows Start Menu**. Click **Autopsy 4.20.0**.



2. Click **OK**.



3. On the welcome screen, select **New Case**.



4. Enter **Analysis 1** for **Case Name**. For Base Directory, browse to select **Desktop**. Click **Next**.

The screenshot shows the 'New Case Information' dialog box with the 'Case Information' tab selected. The 'Steps' list on the left shows '1. Case Information' as the current step. The 'Case Name' field contains 'Analysis 1'. The 'Base Directory' field contains 'C:\Users\Administrator\Desktop\'. A 'Browse' button is next to the 'Base Directory' field. The 'Case Type' section has 'Single-User' selected with a radio button. Below this, a text box indicates 'Case data will be stored in the following directory:' followed by 'C:\Users\Administrator\Desktop\Analysis 1'. At the bottom, there are navigation buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'. The 'Next >' button is highlighted with a red border.

New Case Information

Steps

1. Case Information
2. Optional Information

Case Information

Case Name: Analysis 1

Base Directory: C:\Users\Administrator\Desktop\ Browse

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

C:\Users\Administrator\Desktop\Analysis 1

< Back Next > Finish Cancel Help

5. For **Number**, enter **001** . For **Name**, enter **Jay Doe**. Click **Finish**.

The screenshot shows the 'New Case Information' dialog box with the 'Optional Information' tab selected. The 'Steps' list on the left shows '2. Optional Information' as the current step. The 'Case' section has a 'Number' field containing '001'. The 'Examiner' section has a 'Name' field containing 'Jay Doe', and empty fields for 'Phone', 'Email', and 'Notes'. The 'Organization' section has a dropdown menu set to 'Not Specified' and a 'Manage Organizations' button. At the bottom, there are navigation buttons: '< Back', 'Next >', 'Finish', 'Cancel', and 'Help'. The 'Finish' button is highlighted with a red border.

New Case Information

Steps

1. Case Information
2. Optional Information

Optional Information

Case

Number: 001

Examiner

Name: Jay Doe

Phone:

Email:

Notes:

Organization

Organization analysis is being done for: Not Specified Manage Organizations

< Back Next > Finish Cancel Help

6. Select **Generate new host name based on data source name**. Click **Next**.

The screenshot shows a wizard window titled "Add Data Source" with a close button (X) in the top right corner. On the left, a "Steps" sidebar lists five steps: 1. Select Host (highlighted), 2. Select Data Source Type, 3. Select Data Source, 4. Configure Ingest, and 5. Add Data Source. The main area is titled "Select Host" and contains the text "Hosts are used to organize data sources and other data." Below this text are three radio button options: "Generate new host name based on data source name" (which is selected and has a dashed border), "Specify new host name" (with an adjacent text input field), and "Use existing host" (with an adjacent large empty rectangular box). At the bottom of the window, there are five buttons: "< Back", "Next >" (highlighted with a blue border), "Finish", "Cancel", and "Help".

Add Data Source

Steps

- Select Host**
- Select Data Source Type
- Select Data Source
- Configure Ingest
- Add Data Source

Select Host

Hosts are used to organize data sources and other data.

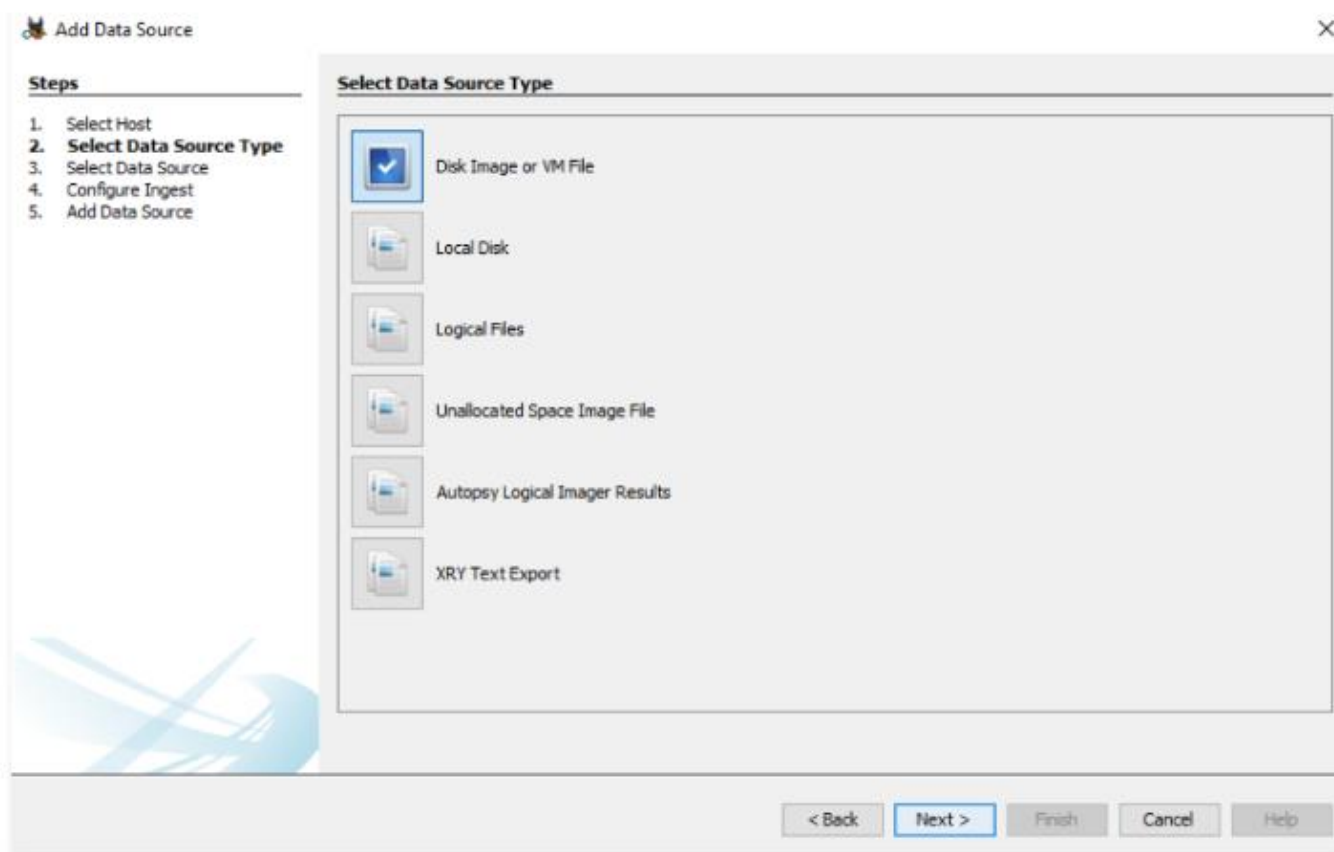
☒ Generate new host name based on data source name

☐ Specify new host name

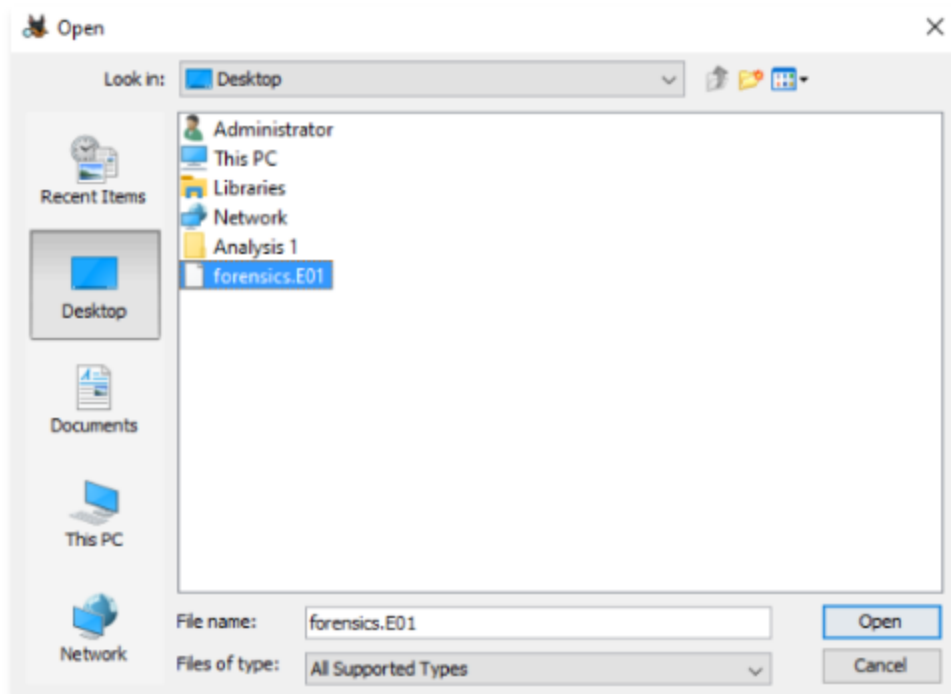
☐ Use existing host

< Back **Next >** Finish Cancel Help

7. Select **Disk Image or VM File**. Click **Next**.



8. For the data source **Path**, select **Browse**. Select **Desktop** in the left pane and open **forensics.E01**.



9. Click **Next**.

Add Data Source

Steps

1. Select Host

2. Select Data Source Type

3. **Select Data Source**

4. Configure Ingest

5. Add Data Source

Select Data Source

Path:

C:\Users\Administrator\Desktop\forensics.E01

Browse

☐ Ignore orphan files in FAT file systems

Time zone:

(GMT+0:00) UTC

Sector size:

Auto Detect

Hash Values (optional):

MD5:

SHA-1:

SHA-256:

NOTE: These values will not be validated when the data source is added.

< Back

Next >

Finish

Cancel

Help

10. Click **Next**.

The screenshot shows the 'Add Data Source' wizard in a software application. The window has a title bar with a close button (X) in the top right corner. On the left side, there is a 'Steps' panel with a list of five steps: 1. Select Host, 2. Select Data Source Type, 3. Select Data Source, 4. **Configure Ingest** (highlighted), and 5. Add Data Source. The main area of the window is titled 'Configure Ingest'. It features a section 'Run ingest modules on:' with a dropdown menu currently set to 'All Files, Directories, and Unallocated Space'. Below this is a list of ingest modules, each with a checked checkbox: Recent Activity (highlighted in blue), Hash Lookup, File Type Identification, Extension Mismatch Detector, Embedded File Extractor, Picture Analyzer, Keyword Search, Email Parser, Encryption Detection, Interesting Files Identifier, Central Repository, PhotoRec Carver, Virtual Machine Extractor, and Data Source Integrity. At the bottom of this list are three buttons: 'Select All', 'Deselect All', and 'History'. To the right of the module list is a text box that says 'The selected module has no per-run settings.' Below this text box is a description for the 'Recent Activity' module: 'Extracts recent user activity, such as Web browsing, recently us...'. A 'Global Settings' button is located at the bottom right of this section. At the very bottom of the window, there is a navigation bar with four buttons: '< Back', 'Next >' (highlighted with a blue border), 'Finish', 'Cancel', and 'Help'.

Add Data Source

Steps

1. Select Host
2. Select Data Source Type
3. Select Data Source
- 4. Configure Ingest**
5. Add Data Source

Configure Ingest

Run ingest modules on:

All Files, Directories, and Unallocated Space

- ☒ Recent Activity
- ☒ Hash Lookup
- ☒ File Type Identification
- ☒ Extension Mismatch Detector
- ☒ Embedded File Extractor
- ☒ Picture Analyzer
- ☒ Keyword Search
- ☒ Email Parser
- ☒ Encryption Detection
- ☒ Interesting Files Identifier
- ☒ Central Repository
- ☒ PhotoRec Carver
- ☒ Virtual Machine Extractor
- ☒ Data Source Integrity

Select All Deselect All History

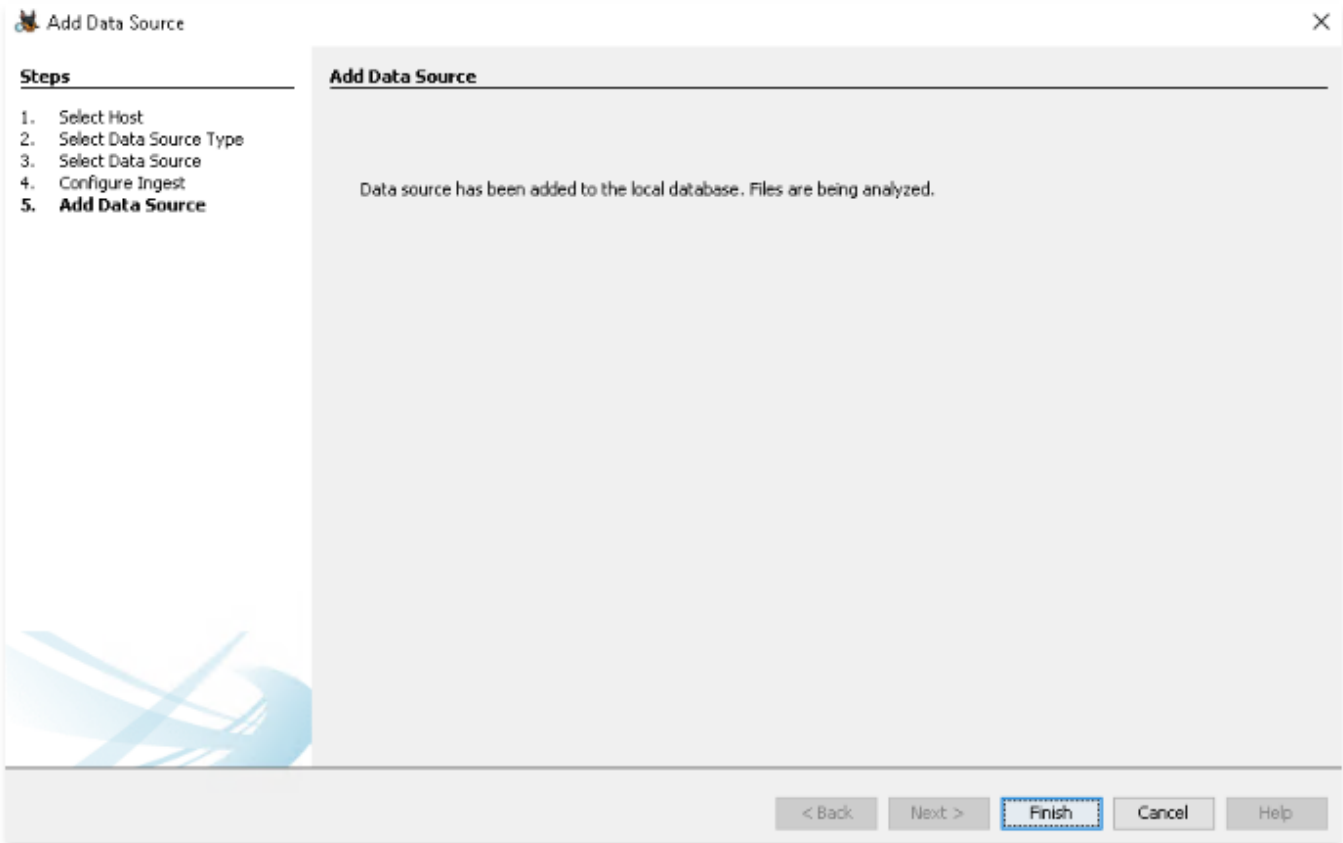
The selected module has no per-run settings.

Extracts recent user activity, such as Web browsing, recently us...

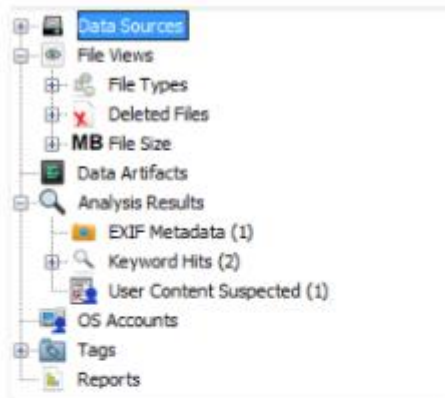
Global Settings

< Back **Next >** Finish Cancel Help

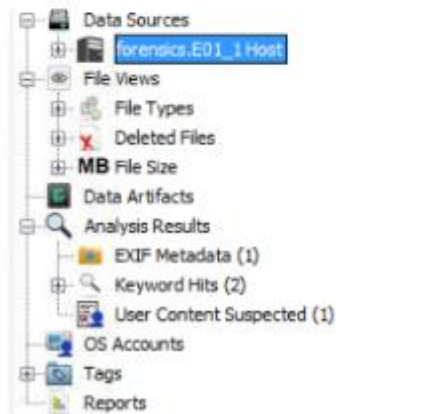
11. Click **Finish**.



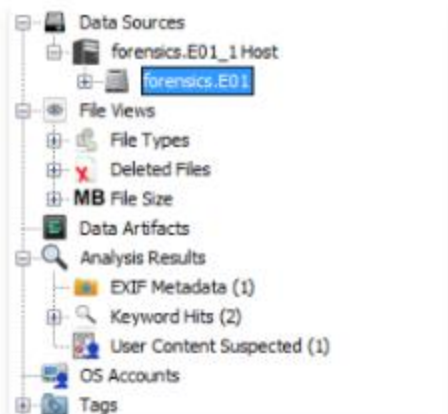
12. Expand **Data Sources** in the left pane.



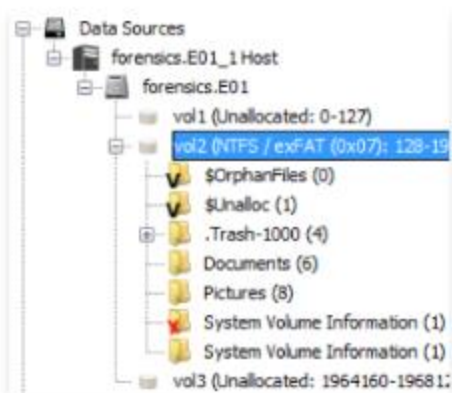
13. Click **forensics.E01_1 Host**.



14. Click **forensics.E01**.



15. Click **vol2**.



16. Select **Documents**. Right click **secret.txt** and select **Extract File(s)**. Save to desktop as **secret.txt**.

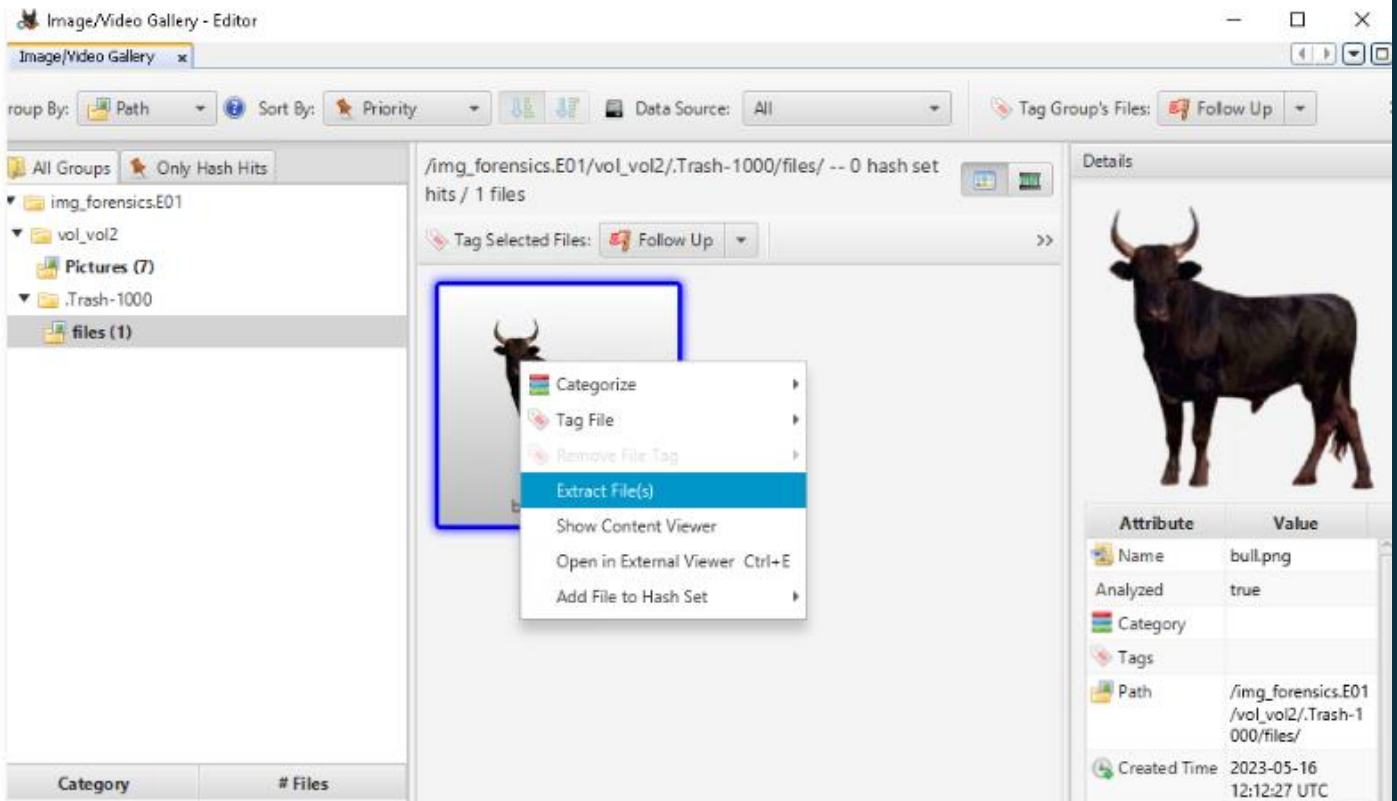
The screenshot displays a forensic analysis tool interface. On the left, a tree view shows the file system structure under 'Data Sources'. The 'Documents' folder is selected. The main pane shows a list of files, including 'secret.txt'. A right-click context menu is open over 'secret.txt', with the 'Extract File(s)' option highlighted. The bottom of the interface shows a table with columns for Name, S, C, O, Modified Time, Change Time, and Access Time. The table lists several files, including 'secret.txt'.

Name	S	C	O	Modified Time	Change Time	Access Time
.secret.txt.kate-swp				2023-05-16 12:35:50 UTC	0000-00-00 00:00:00	2023-05-16 12:35:38
goodbye.txt			0	2023-05-15 14:34:58 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28
hello.txt			0	2023-05-15 14:35:04 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28
numbers.txt			0	2023-05-15 14:34:04 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28
secret.txt				2023-05-16 12:35:51 UTC	0000-00-00 00:00:00	2023-05-16 12:35:52
work.txt				2023-05-15 14:34:42 UTC	0000-00-00 00:00:00	2023-05-16 05:12:28

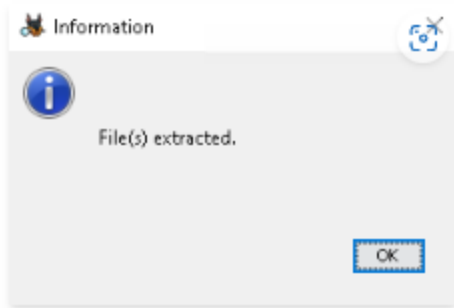
17. Click Images/Videos at the top.



18. Select **img_forensics.E01** → **vol_vol2** → **.Trash-1000** → **files**. Right click **bull.png** and **Extract File(s)**. Save to desktop as **bull.png**.



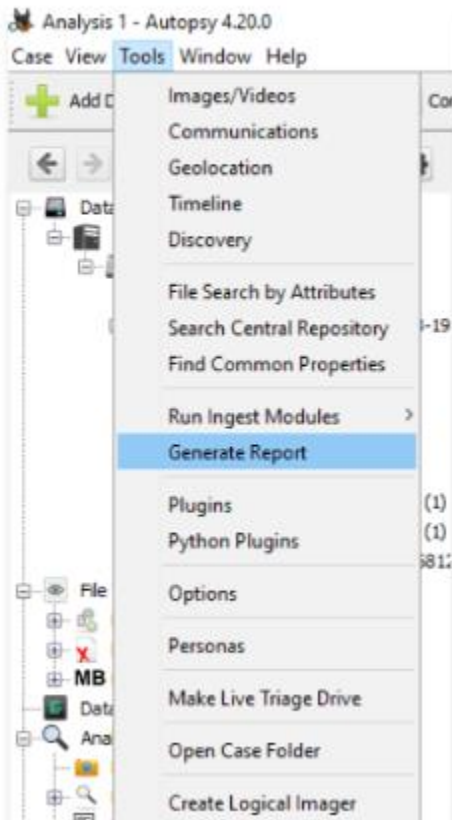
19. Click **OK** on the popup.



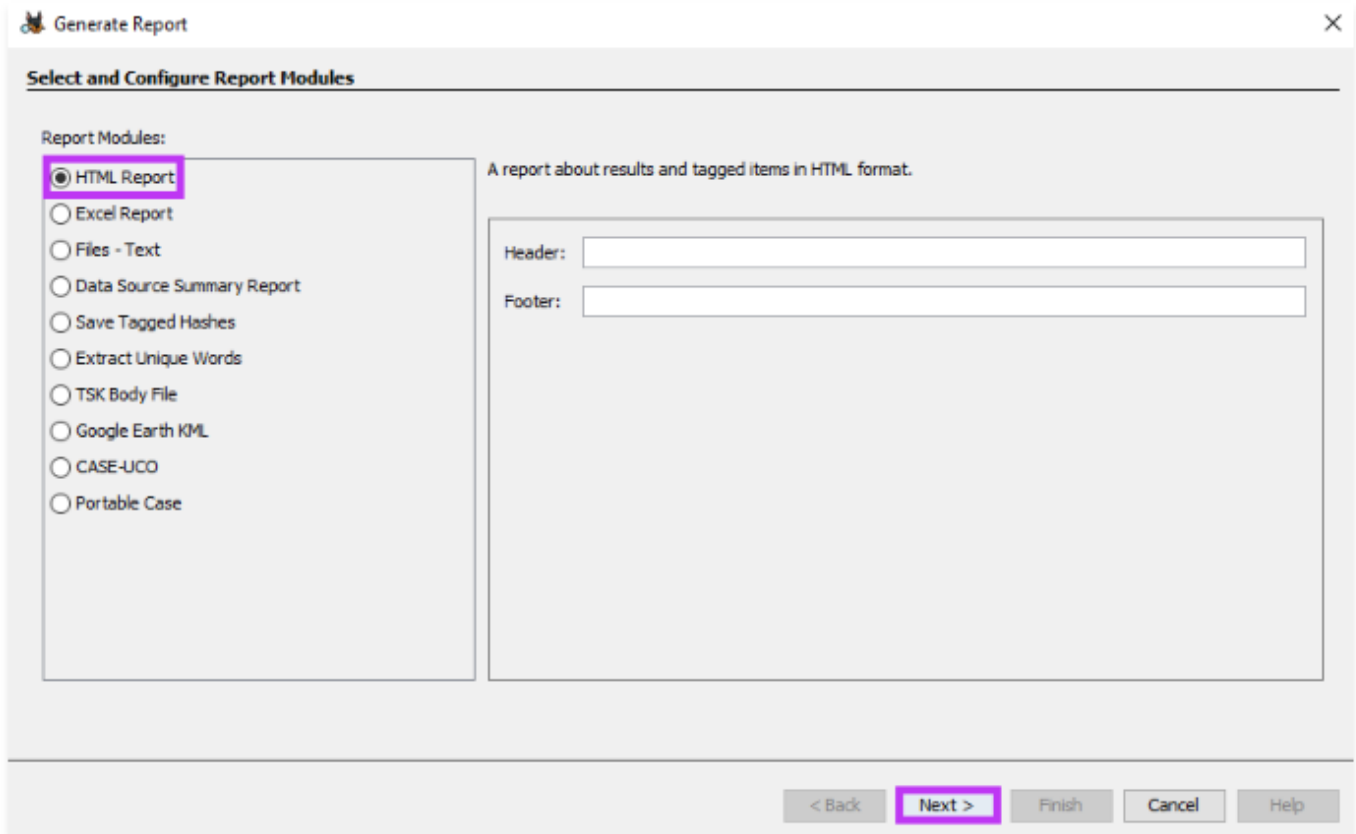
20. Close the **Image/Video Gallery - Editor**.

Task 3: Generate a report with Autopsy

1. At the top, select **Tools** and then **Generate Report**.



2. Select **HTML report**. Click **Next**.



The image shows a 'Generate Report' dialog box with a title bar containing a cat icon and a close button. The main area is titled 'Select and Configure Report Modules'. On the left, under 'Report Modules:', there is a list of radio buttons. The first option, 'HTML Report', is selected and highlighted with a red rectangle. Below it are 'Excel Report', 'Files - Text', 'Data Source Summary Report', 'Save Tagged Hashes', 'Extract Unique Words', 'TSK Body File', 'Google Earth KML', 'CASE-UCO', and 'Portable Case'. To the right of this list, there is a description: 'A report about results and tagged items in HTML format.' Below this description are two text input fields labeled 'Header:' and 'Footer:'. At the bottom of the dialog, there is a row of five buttons: '< Back', 'Next >' (highlighted with a red rectangle), 'Finish', 'Cancel', and 'Help'.

Generate Report

Select and Configure Report Modules

Report Modules:

- ☒ HTML Report
- ☐ Excel Report
- ☐ Files - Text
- ☐ Data Source Summary Report
- ☐ Save Tagged Hashes
- ☐ Extract Unique Words
- ☐ TSK Body File
- ☐ Google Earth KML
- ☐ CASE-UCO
- ☐ Portable Case

A report about results and tagged items in HTML format.

Header:

Footer:

< Back Next > Finish Cancel Help

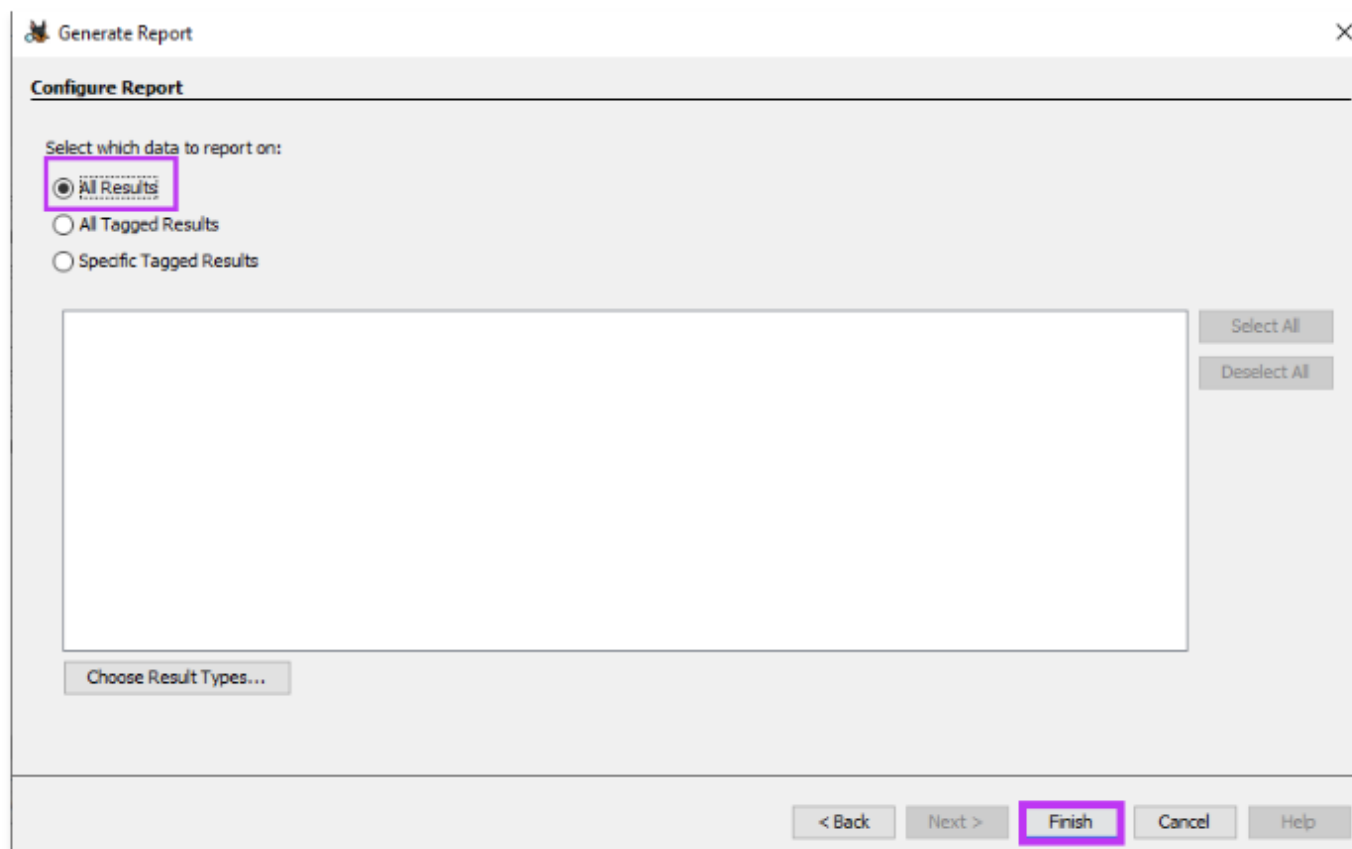
3. Check **forensics.E01**. Click **Next**.

Generate Report ×

Select which data source(s) to include

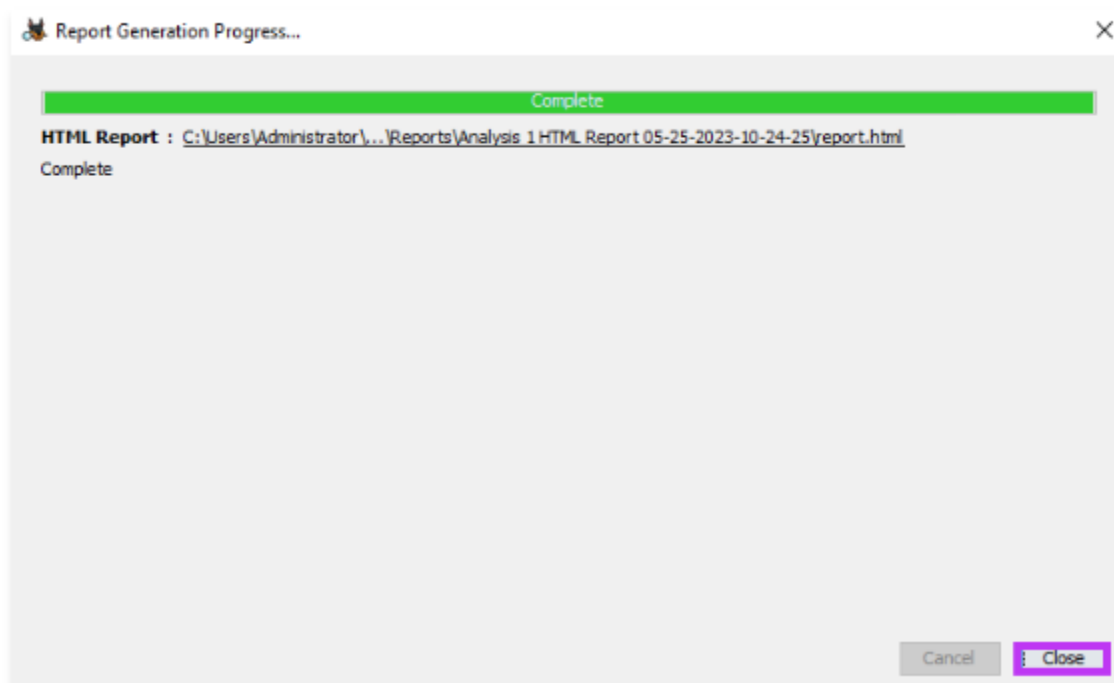
☒ forensics.E01

4. Select **All Results**. Click **Finish**.



The 'Generate Report' dialog box is titled 'Generate Report' with a close button (X) in the top right corner. Below the title bar is a section labeled 'Configure Report'. Inside this section, the text 'Select which data to report on:' is followed by three radio button options: 'All Results' (which is selected and highlighted with a red rectangle), 'All Tagged Results', and 'Specific Tagged Results'. Below these options is a large empty rectangular box. To the right of this box are two buttons: 'Select All' and 'Deselect All'. Below the large box is a button labeled 'Choose Result Types...'. At the bottom of the dialog box, there is a row of five buttons: '< Back', 'Next >', 'Finish' (highlighted with a red rectangle), 'Cancel', and 'Help'.

5. Click **Close**.



The 'Report Generation Progress' dialog box is titled 'Report Generation Progress...' with a close button (X) in the top right corner. It features a green progress bar at the top, which is fully filled and labeled 'Complete'. Below the progress bar, the text 'HTML Report : C:\Users\Administrator\...\Reports\Analysis 1 HTML Report 05-25-2023-10-24-25\report.html' is displayed, followed by the word 'Complete'. At the bottom right of the dialog box, there are two buttons: 'Cancel' and 'Close' (highlighted with a red rectangle).

6. Go to the location the report is saved in. Double click the report.html link to view the autopsy report.

Report Navigation

- Case Summary
- ★ Data Source Usage (1)
- 📷 EXIF Metadata (1)
- 🔍 Keyword Hits (2)
- ★ Tagged Files (0)
- ★ Tagged Images (0)
- ★ Tagged Results (0)
- 🔍 User Content Suspected (1)

Autopsy Forensic Report

HTML Report Generated on 2023/07/06 19:20:25

Case:	Analysis 1
Case Number:	001
Number of data sources in case:	1
Examiner:	Jay Doe

Image Information:

forensics.E01

Timezone:	UTC
Path:	C:\Users\Administrator\Desktop\forensics.E01

Software Information:

Autopsy Version:	4.20.0
Android Analyzer Module:	4.20.0
Android Analyzer (aLEAPP) Module:	4.20.0

7. The lab is now complete. To end the lab, click **Delete VM** in the right pane.